

Toe Physics

So this blog is a bit different from the normal game update and it's actually been something I've been slowly chipping at for a few months now! So back when I wrote my review on The Muscle Ladder I actually picked up a second book to read on the side: Neumann's Kinesiology. Since this book is catered towards graduate studies I initially didn't bother with any blog-related work; I mostly took notes for myself as I slowly went through the textbook. However as a firm believer of the Feynman method of learning I figured it was worth paraphrasing my notes and learnings in a blog shaped Trojan Horse. Specifically we will be covering Chapter 14 which focuses on how the feet and legs come together to enable gait.

However I do think that there should be some central theming outside of the body part itself so this blog will be dedicated to some of the more interesting mechanics that I personally didn't know!

Carpenter's Mortise

Very similar to a carpenter joint and serves to allow the ankle to shift between two major positions - pronated and supinated (pointing up and down). Of course this is a delicate balance primarily handled by the ligaments. If you wanted to limit this joint then you could free-fall to your knees while keeping your feet planted on the ground. This action basically makes your ligaments explode so I wouldn't recommend trying this at home. The reverse example of a toe stealer popping from the ground and horrifically yanking your toes into the ground would directly damage the bones.

Windlass Effect

From an engineering perspective a Windlass is pretty generic since it's a rope which is pulled upon (either from a lever or pulley) and subsequently applies force on the desired object. Some examples include a tourniquet and a double pulley system.

Within the context of the human foot the Windlass comprises the big toe, the plantar fascia, and the calcaneus. The big toe will extend (usually via a kick-off during gait) which causes the anchored plantar fascia to become taut, allowing the single foot to bear the weight of a person at the peak of their walking cycle.

Second Class Lever

This is pretty rare in the human body (despite having an entire blog on weird physics quirks). Specifically the interesting part is that the body normally uses 1:1 leverage but the calf + foot have a 3:1 ratio which allows your relatively small calves to support the body when walking/running/tip-toeing. For those who somehow remember levers from their physics classes then this mimics a wheelbarrow where the calves “pick” the weight up from the ground by the pivot point known as the toes.

Tarsal Tunnel Syndrome

This medical condition is identical to Carpal Tunnel Syndrome in the sense that a nerve (the Tarsal Nerve) is entrapped (compressed) by its surrounding inflamed tendons/ligaments etc. Located in between the Achilles Tendon and the exterior Talocrural Joint, this doesn't seem like a common ailment when compared to Carpal Tunnel due to most corporate jobs requiring repetitive movements in the wrists.

Davis' Law

All tendons/ligaments get strangers as you apply more and more constant stress. The main problem with confirming this is that tendons and ligaments don't get visibly thicker or denser like bones. We're also not allowed to stretch people until things snap... but in the real world this is easily observable with power lifting, extreme sports, etc

Wolff's Law

It's basically Davis' Law but with bones (AKA strength training increases bone density) and it's a LOT easier to measure without literally snapping tendons/ligaments. This also goes in the other direction and is the reason why drugs like GLP-1 inhibitors are so dangerous to sedentary folks. They don't change their lifestyle and suddenly find that their bones will just shatter for fun!

Flat Foot Deformity

Basically messes with the Windlass effect creating a litany of issues: More force needed to kick-off, horrifically unstable ankle, and over-exertion of every other foot muscle increasing the risk for more injuries.

Hallux Valgus Deformity

While most people call these bunions, big toe deformities can vary (mostly from the root cause) and all have major impact on both gait and balance. More specifically the big toe lacks the interstitial ligaments that the other toes have (between the phalanges) and thus has a separate purpose - to drive the foot forward during any kick-off and by extension bear the brunt of the force.

Final Notes

Once again, these notes are mostly for me to review and actively think about the material I've learned from the textbook so if anything feels inaccurate/confusing/incomplete then whoops! If there's any consolation then we will likely resume our regularly scheduled gaming posts (unless I get a banger idea in-between today and next blog).